

Universidad Peruana de Ciencias Aplicadas



MIDTERM PROJECT REPORT

Big Data

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Computer Science

Section:

18519

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Table of Contents

Introduction.....	4
Problem Statement.....	4
Product Question.....	4
Source Inventory.....	4
Schema Draft.....	4
Entity Tables.....	5
Keys (Primary and Foreign Keys).....	5
Relationships and Joins.....	6
ERD.....	7
Data Dictionary.....	7
Raw Dataset Overview.....	7
Tables Description.....	9
Ethics and Access Note.....	12
Preprocessing and feature engineering.....	13
Preprocessing.....	13
Data Loading and Initial Inspection.....	13
Duplicate Handling.....	13
Variable Identification.....	14
Missing Values Check.....	14
Date Column Conversion.....	15
Scale Analysis.....	15
Feature Engineering.....	16
patient_feature_matrix:.....	16
Source Tables:.....	16
Join Operation:.....	16
Aggregation Level:.....	16
Features Included:.....	17
Data Types.....	17
Dimensions:.....	17
Feature Matrix Expansion.....	19
Dimensionality and Representation Analysis.....	20
Motivation.....	20
Methods.....	20
PCA.....	20
t-SNE.....	22
Data Preparation.....	23
Results.....	24
Explained Variance.....	24
Component Selection.....	24
Comparison Table.....	24

Visualizations.....	25
K-Means Experiment.....	29
K Selection.....	29
K-Means Model.....	32
Model Training.....	34
Cluster Visualization.....	34
DBSCAN Experiment.....	36
Parameter Selection.....	36
DBSCAN Parameter Sweep Analysis.....	37
Validation Table.....	38
Cluster Profile Analysis.....	39
Cluster 0: "Standard High-Turnover Care Group A" (Low-Risk Operational Baseline). 40	
Cluster 1: "Extended Stay / Low-Intensity Segment" (Operational Bed-Occupancy Bottleneck).....	41
Cluster 2: "High-Intensity & High-Risk Critical Segment" (Maximum Exposure & Margin Risk).....	42
Cluster 3: "Standard High-Turnover Care Group B" (Symmetric Commercial Runtime).....	43
Failure Analysis.....	44
a. K-Means Structural Boundary Overlap.....	44
b. DBSCAN Density Disruption & Outlier Value.....	44
Technical Interpretation.....	45
1. Dimensionality Reduction & Latent Space Structure (PCA Analysis).....	45
2. Clustering Topology & Mathematical Validation.....	46
K-Means Convergence & Compactness.....	46
DBSCAN Density Collapse.....	47
3. Algorithmic Trade-Off & Final Production Selection.....	48
4. Model Validation Summary Table.....	48
Recommendation, Ranking, or Predictive Decision Engine.....	49
1. Task Framing and Product Context.....	49
2. Data Alignment and Experiment Design.....	50
3. Offline Evaluation Protocol.....	52
4. Baseline System Implementation.....	54
5. Stronger System Development.....	55
6. Comparative Performance Report.....	56
7. Error Analysis and Case Studies.....	59
Conclusions.....	62
Anexos.....	65

Introduction

This project focuses on building a data intelligence pipeline for the healthcare sector. Using a synthetic dataset of 55,500 patient records, we aim to implement a complete lifecycle from automated ingestion to advanced entity clustering and graph-based medical analysis.

Problem Statement

The core challenge is the lack of structured, high-integrity data in hospital management. This project addresses the need for a "Catalog Layer" that cleans raw data, standardizes medical categories, and handles high-cardinality features, providing a foundation for predictive clinical insights.

Product Question

Is it possible to cluster patient profiles and treatment pathways to identify patterns between medical conditions, medication types, and hospital admission outcomes?

Source Inventory

- **Primary Source:** healthcare_dataset.csv (Synthetic Healthcare Data).
- **Entity Count:** Patients (unique individuals), Doctors, and Healthcare Providers.
- **Key Features:** Medical Condition, Admission Type, Medication, and Test Results.
- **Temporal Scope:** Continuous tracking from Admission Date to Discharge Date.

Dataset url	https://www.kaggle.com/datasets/prasad22/healthcare-dataset
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Schema Draft

Entity Tables

The data model is composed of multiple entity tables derived from the original dataset. Each table represents a key component of the healthcare domain and is designed to support relational and analytical operations.

The main entities include:

- **patients**: stores demographic information about each patient, such as age, gender, and blood type.
- **hospitals**: contains identifiers and names of healthcare institutions.
- **doctors**: includes information about attending medical professionals.
- **medical_conditions**: represents diagnosed health conditions.
- **medications**: represents prescribed treatments.
- **medical_records**: serves as the central table that captures each clinical event, including patient interactions, diagnosis, treatment, and hospital-related data.

Keys (Primary and Foreign Keys)

The schema uses primary keys (PK) to uniquely identify each record within a table, and foreign keys (FK) to establish relationships between tables.

Primary Keys (PK):

- patients.patient_id
- hospitals.hospital_id
- doctors.doctor_id
- medical_conditions.condition_id
- medications.medication_id

- `medical_records.record_id`

Foreign Keys (FK):

- `medical_records.patient_id` → `patients.patient_id`
- `medical_records.hospital_id` → `hospitals.hospital_id`
- `medical_records.doctor_id` → `doctors.doctor_id`
- `medical_records.condition_id` → `medical_conditions.condition_id`
- `medical_records.medication_id` → `medications.medication_id`

Relationships and Joins

The schema defines one-to-many relationships between entity tables and the central **medical_records** table.

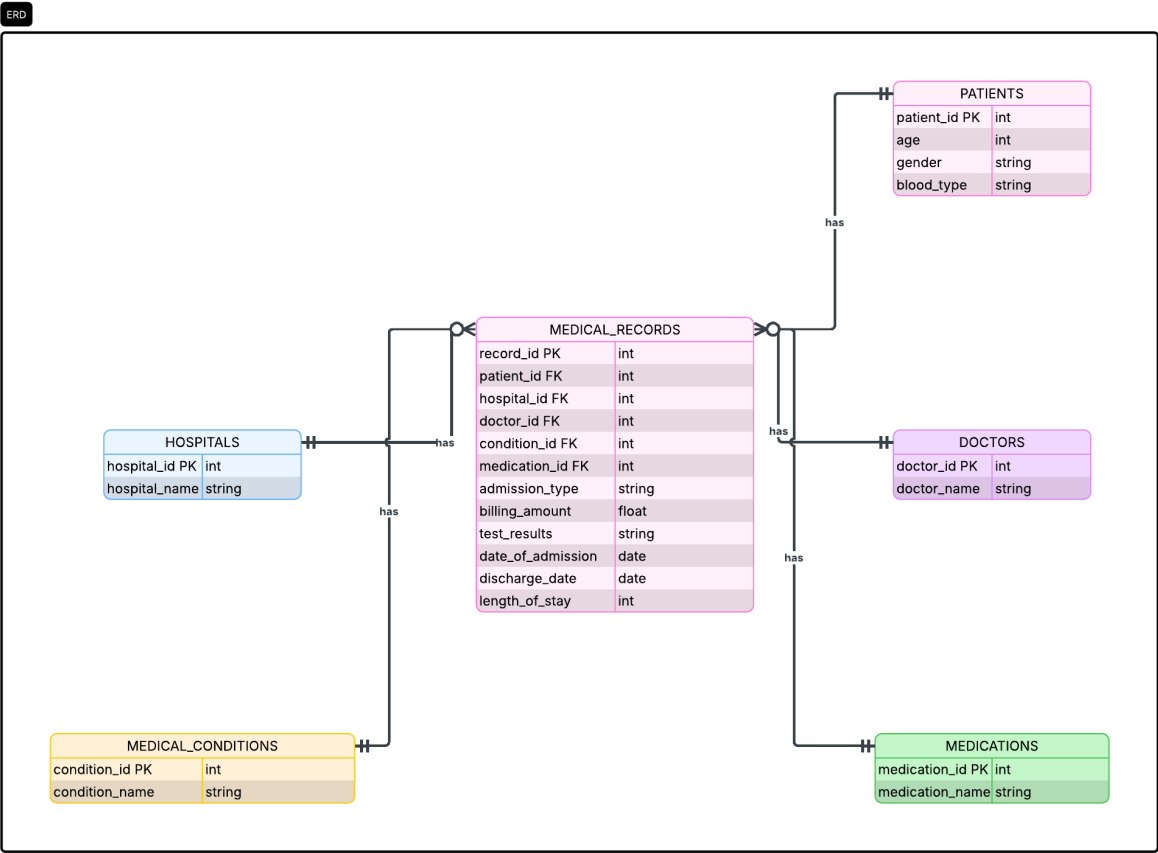
- One patient can have multiple medical records.
- One hospital can have multiple medical records.
- One doctor can be associated with multiple medical records.
- One medical condition can appear in multiple records.
- One medication can appear in multiple records.

These relationships are implemented through foreign keys and enable efficient joins for analytical purposes.

Example joins include:

- `patients ↔ medical_records` (via `patient_id`)
- `hospitals ↔ medical_records` (via `hospital_id`)
- `medical_conditions ↔ medical_records` (via `condition_id`)
- `medications ↔ medical_records` (via `medication_id`)

ERD



The schema transforms the original dataset into a normalized relational model centered on medical_records. Key entities such as patients, hospitals, doctors, conditions, and medications are separated into individual tables and connected through foreign keys, enabling scalable analysis, clustering, and graph-based modeling.

Data Dictionary

Raw Dataset Overview

Column Name	Description	Data Type
-------------	-------------	-----------

Name	Patient's name	string
Age	Age of the patient	integer
Gender	Gender of the patient (Male, Female)	categorical
Blood Type	Blood type of the patient	categorical
Medical Condition	Diagnosed medical condition	categorical
Date of Admission	Date of patient admission	datetime
Doctor	Attending doctor's name	string
Hospital	Hospital name	string
Insurance Provider	Insurance company	categorical
Billing Amount	Total billing amount	float
Room Number	Room assigned to the patient	integer
Admission Type	Type of admission (Emergency, Elective, Urgent)	categorical
Discharge Date	Date of patient discharge	datetime
Medication	Prescribed medication	categorical
Test Results	Results of medical tests	categorical

The raw dataset was transformed into multiple normalized tables to support relational analysis, clustering, and graph-based modeling.

Tables Description

Table: Patients

The patients table contains demographic information for each individual in the dataset. It includes attributes such as age, gender, and blood type, and each patient is uniquely identified by a patient_id. This table serves as a core entity for analyzing patient-level patterns.

Column	Type	Description
patient_id	int	unique identifier for each patient
age	int	age of the patient
gender	categorical	patient gender
blood_type	categorical	blood type of the patient

Table: Hospitals

The hospitals table stores information about healthcare institutions where patients receive treatment. Each hospital is uniquely identified by a hospital_id and is linked to medical records through relational keys.

Column	Type	Description
hospital_id	int	unique identifier for each hospital

hospital_name	string	name of the hospital
----------------------	--------	----------------------

Table: Doctors

The doctors table contains information about medical professionals responsible for patient care. Each doctor is uniquely identified by a doctor_id and may be associated with multiple medical records.

Column	Type	Description
doctor_id	int	unique identifier for each doctor
doctor_name	string	name of the doctor

Table: medical_conditions

The medical_conditions table represents the set of diagnosed health conditions. Each condition is assigned a unique condition_id, enabling consistent referencing and reducing redundancy across the dataset.

Column	Type	Description
condition_id	int	unique identifier for the condition
condition_name	categorical	name of the medical condition

Table: Medications

The medications table includes the different treatments or drugs prescribed to patients. Each medication is uniquely identified by a medication_id and is linked to medical records for analysis of treatment patterns.

Column	Type	Description
medication_id	int	unique identifier for the medication
medication_name	ategorical	name of the medication

Table: medical_records

The medical_records table is the central fact table of the dataset. It captures each clinical event, including patient interactions with hospitals and doctors, diagnoses, treatments, admission details, and billing information. This table connects all other entities through foreign keys and enables comprehensive analytical tasks such as clustering, pattern discovery, and graph-based modeling.

Column	Type	Description
record_id	int	unique identifier for each record
patient_id	int	reference to patients table

hospital_id	int	reference to hospitals table
doctor_id	int	reference to doctors table
condition_id	int	reference to medical_conditions table
medication_id	int	reference to medications table
admission_type	categorical	type of admission (Emergency, Elective, Urgent)
billing_amount	float	total cost of treatment
test_results	categorical	results of medical tests
date_of_admission	datetime	admission date
discharge_date	datetime	discharge date
length_of_stay	int	number of days hospitalized

Ethics and Access Note

- **Data Origin:** The data is a synthetic dataset designed for educational and data science research.
- **Usage Justification:** We are utilizing this dataset under academic license for the "Domain Discovery, Recommendation, and Graph Intelligence" course.
- **Personal Data Risks and Mitigation:** While the data contains "Name" fields, they are synthetically generated and do not correspond to real individuals (PII). To prevent any

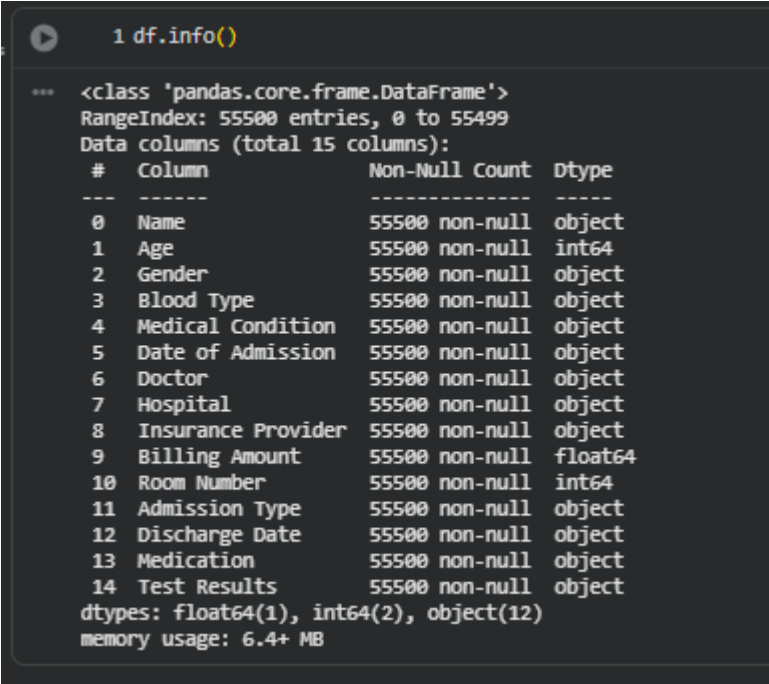
accidental correlation, these identifiers will be treated as non-functional tokens or excluded from the final feature vectors.

Preprocessing and feature engineering

Preprocessing

Data Loading and Initial Inspection

The dataset, `healthcare_dataset.csv`, was loaded into a Pandas DataFrame. An initial inspection using `df.info()` revealed 15 columns with various data types, primarily object (strings), int, and float.



```
1 df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 55500 entries, 0 to 55499
Data columns (total 15 columns):
 #   Column                Non-Null Count  Dtype  
---  -
 0   Name                  55500 non-null  object  
 1   Age                   55500 non-null  int64   
 2   Gender                55500 non-null  object  
 3   Blood Type            55500 non-null  object  
 4   Medical Condition     55500 non-null  object  
 5   Date of Admission     55500 non-null  object  
 6   Doctor                55500 non-null  object  
 7   Hospital              55500 non-null  object  
 8   Insurance Provider    55500 non-null  object  
 9   Billing Amount         55500 non-null  float64  
10  Room Number           55500 non-null  int64   
11  Admission Type        55500 non-null  object  
12  Discharge Date        55500 non-null  object  
13  Medication            55500 non-null  object  
14  Test Results          55500 non-null  object  
dtypes: float64(1), int64(2), object(12)
memory usage: 6.4+ MB
```

Duplicate Handling

A check for duplicate rows identified 534 duplicate entries. These duplicates were subsequently removed using `df.drop_duplicates(inplace=True)`, resulting in a cleaned dataset of 54,966 unique rows.

```
1 df.drop_duplicates(inplace=True)
2 print(f"Number of rows after removing duplicates: {len(df)}")

Number of rows after removing duplicates: 54966
```

Variable Identification

Columns were categorized into numerical and categorical types:

- Categorical Columns: Name, Gender, Blood Type, Medical Condition, Date of Admission, Doctor, Hospital, Insurance Provider, Admission Type, Discharge Date, Medication, Test Results.
- Numerical Columns: Age, Billing Amount, Room Number.

```
1 categorical_cols = df.select_dtypes(include='object').columns
2 numerical_cols = df.select_dtypes(include=['int64', 'float64']).columns
3
4 print("Categorical columns:", categorical_cols.tolist())
5 print("Numerical columns:", numerical_cols.tolist())

Categorical Columns: ['Name', 'Gender', 'Blood Type', 'Medical Condition', 'Date of Admission', 'Doctor', 'Hospital', 'Insurance Provider', 'Admission Type', 'Discharge Date', 'Medication', 'Test Results']
Numerical Columns: ['Age', 'Billing Amount', 'Room Number']
```

Missing Values Check

No missing values were found across any of the columns in the dataset, eliminating the need for imputation.

```
1 print('Missing values per column:')
2 print(df.isnull().sum())

*** Missing values per column:
Name          0
Age           0
Gender        0
Blood Type    0
Medical Condition 0
Date of Admission 0
Doctor        0
Hospital      0
Insurance Provider 0
Billing Amount 0
Room Number   0
Admission Type 0
Discharge Date 0
Medication    0
Test Results  0
dtype: int64
```

Date Column Conversion

The columns Date of Admission and Discharge Date were converted from object (string) to datetime data type using `pd.to_datetime`. This is crucial for any time-based analysis, such as calculating the length of stay.

```
1 df['Date of Admission'] = pd.to_datetime(df['Date of Admission'])
2 df['Discharge Date'] = pd.to_datetime(df['Discharge Date'])
3
4 print('Data types after date conversion:')
5 display(df[['Date of Admission', 'Discharge Date']].info())

Data types after date conversion:
<class 'pandas.core.frame.DataFrame'>
Index: 54966 entries, 0 to 55499
Data columns (total 2 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   Date of Admission  54966 non-null  datetime64[ns]
1   Discharge Date    54966 non-null  datetime64[ns]
dtypes: datetime64[ns](2)
memory usage: 1.3 MB
None
```

Scale Analysis

- Number of Rows and Columns: 55,500 initial records and 15 columns. After removing duplicates, the dataset maintains 54,966 clean records.

- Missingness: The dataset is complete with 0% missing values, ensuring high reliability for statistical modeling.
- Sparsity or Memory Estimation: This is a dense dataset with a memory footprint of approximately 6.4 MB, allowing for high-performance processing in standard environments.

Feature Engineering

patient_feature_matrix:

Aggregated patient-level feature matrix constructed by combining demographic information with medical records. Each row represents a unique patient and summarizes their healthcare activity and characteristics.

Source Tables:

- patients
- medical_records

Join Operation:

The datasets were merged using a left join on patient_id:

```
# Unir `patients` con `medical_records`  
patient_medical_data = pd.merge(  
    patients,  
    medical_records,  
    on='patient_id',  
    how='left'  
)
```

Aggregation Level:

- Grouped by: patient_id

- Each row represents: one patient

Features Included:

patient_id	Unique patient identifier	—
age	Patient age	first
gender	Patient gender	first
blood_type	Patient blood type	first
total_admissions	Total number of admissions	count
avg_billing_amount	Average billing amount per admission	mean
avg_length_of_stay	Average hospital stay duration (days)	mean

Data Types

- Numeric: age, total_admissions, avg_billing_amount, avg_length_of_stay.
- Categorical: gender, blood_type.

Dimensions:

The feature matrix is structured so that each row represents a single patient, and each column represents a characteristic or summary of that patient's medical history.

In this case, the matrix includes:

- **Patients (rows):** each unique patient_id corresponds to one individual
- **Features (columns):** demographic attributes (such as age, gender, and blood type) and aggregated healthcare metrics (such as total admissions, average billing amount, and average length of stay)

This structure allows us to move from raw, record-level data (multiple rows per patient) to a **patient-centered view**, where all relevant information is summarized into a single row per individual.

As a result:

- The dataset becomes easier to analyze and model
- Each patient can be directly compared to others based on their characteristics
- It serves as a suitable input for machine learning tasks such as clustering, similarity analysis, and recommendation systems

Matriz de Características de Pacientes (Primeras 5 filas):

	patient_id	age	gender	blood_type	total_admissions	avg_billing_amount	avg_length_of_stay
0	0	22	Male	AB-	1	39906.147308	11.0
1	1	76	Female	A+	1	10584.185945	13.0
2	2	20	Male	O-	1	29740.960199	26.0
3	3	69	Female	B-	1	21535.554758	9.0
4	4	18	Male	B-	1	13895.551020	7.0

Dimensiones de la Matriz de Características de Pacientes: (49992, 7)

Feature Matrix Expansion

Motivation (Why the matrix was expanded)

The initial patient feature matrix was expanded to enrich each patient's representation with more descriptive and predictive attributes. The goal was to move beyond basic demographic data and incorporate information about their medical interactions, which is crucial for subsequent tasks such as clustering, segmentation, or building predictive models.

Added Features (List of added features)

Additional features added to the patient matrix included:

1. **One-Hot Encoding** for categorical variables 'Gender' and 'Blood Type'.
2. **Medical Condition Statistics**: Count of each medical condition and the most frequent condition per patient.
3. **Medication Statistics**: Count of each medication and the most frequent medication per patient.
4. **Test Results Statistics**: Proportion of 'Normal' and 'Abnormal' test results per patient.

Resulting Representation (How the final matrix turned out)

The patient feature matrix, after expansion, has 54966 rows (patients) and 33 columns (features), as shown in the code output:

Dimensiones de la Matriz de Características de Pacientes ampliada: (54966, 33)

Matriz de Características de Pacientes ampliada (Primeras 5 filas):

	patient_id	age	gender	blood_type	total_admissions	avg_billing_amount	avg_length_of_stay	Gender_Female	Gender_Male	BloodType_A+	...	most_frequent_condition
0	0	22	Male	AB-	1	39906.147308	11.0	False	True	False	...	Obesity
1	1	76	Female	A+	1	10584.185945	13.0	True	False	True	...	Arthritis
2	2	20	Male	O-	1	29740.960199	26.0	False	True	False	...	Hypertension
3	3	69	Female	B-	1	21535.554758	9.0	True	False	False	...	Diabetes
4	4	18	Male	B-	1	13895.551020	7.0	False	True	False	...	Cancer

5 rows x 33 columns

Dimensionality and Representation Analysis

Motivation

Dimensionality reduction was applied to:

- **High-dimensionality problems:** Reduce the number of features from 33 to a more manageable set.
- **Redundancy:** Eliminate redundant information among features, especially highly correlated ones.
- **Visualization:** Facilitate the representation of data in 2 or 3 dimensions to identify patterns or clusters.
- **Computational Efficiency:** Improve the performance of Machine Learning algorithms by working with a smaller number of dimensions.

Methods

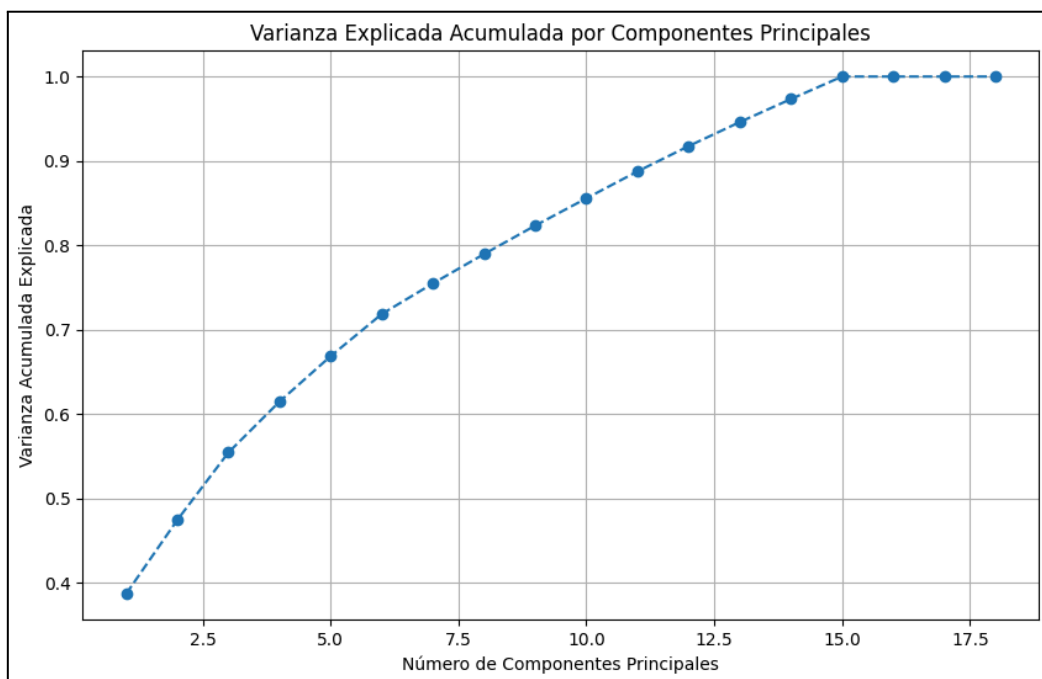
PCA

To reduce the dimensionality of the healthcare dataset, Principal Component Analysis (PCA) was applied after the preprocessing stage. The objective of this technique was to decrease the number of variables while preserving most of the original information contained in the dataset.

The cumulative explained variance graph shows how the amount of retained information increases as more principal components are added. The results indicate that the first 14 principal components preserve approximately 95% of the total variance of the dataset. This means that the dimensionality of the data was significantly reduced while maintaining most of the relevant patterns and relationships between patient records.

The PCA transformation also helped reduce redundancy and noise generated by highly correlated variables and one-hot encoded categorical features. In addition, dimensionality reduction improved the computational efficiency of subsequent techniques such as clustering and t-SNE visualization.

The final transformed dataset contained 1,196 observations represented by 14 principal components. These components summarize the most important patterns related to medical conditions, medications, admission characteristics, billing amounts, and clinical test results.



t-SNE

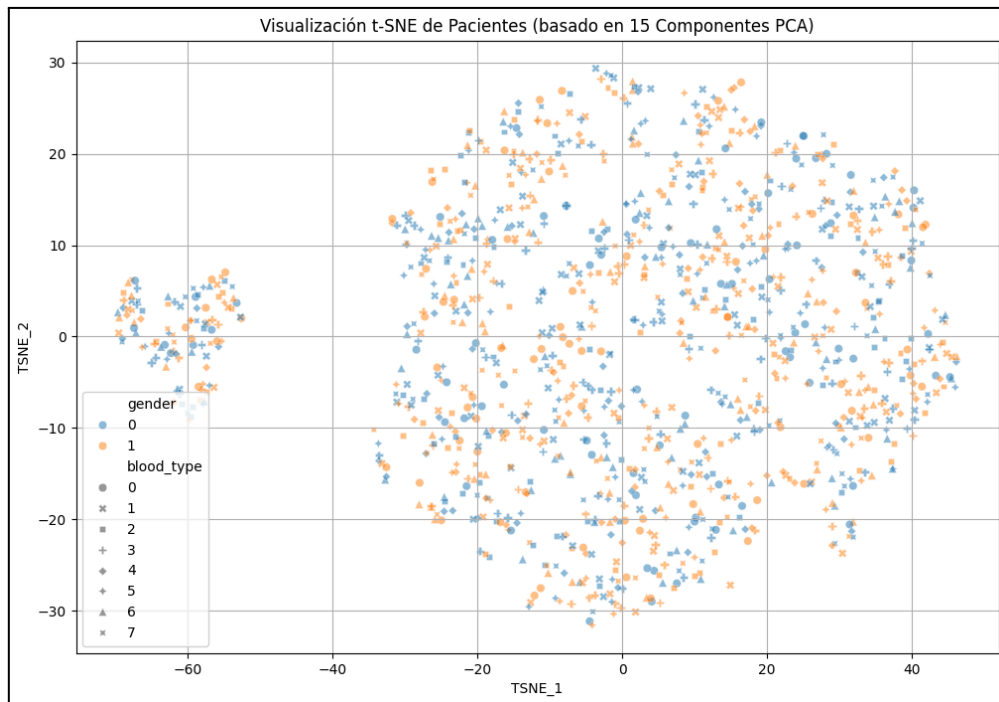
The t-SNE visualization was used to project the high-dimensional patient data into a two-dimensional space in order to identify hidden structures and similarities between patient profiles. The visualization shows a large central distribution of patient records and a smaller separated subgroup located on the left side of the graph.

The main cluster suggests that most patients share relatively similar clinical characteristics, treatment patterns, and hospital admission behaviors. This indicates that the dataset contains continuous relationships between medical conditions, medications, admission types, and test results rather than completely isolated patient groups.

Additionally, a smaller isolated subgroup can be observed around lower TSNE_1 values. This separated region may represent patients with uncommon combinations of medical conditions, treatments, or hospital outcomes. The persistence of this subgroup after preprocessing and dimensionality reduction suggests the presence of meaningful latent patterns within the healthcare dataset.

The visualization also shows that gender and blood type are distributed across the entire embedding space without forming clearly separated regions. This suggests that these variables are not the primary factors driving patient clustering. Instead, the clustering structure appears to be more strongly influenced by clinical and treatment-related variables.

Overall, the t-SNE projection confirms that meaningful patient profile structures exist in the dataset and supports the application of clustering algorithms such as K-Means and DBSCAN for healthcare pattern discovery.



Data Preparation

For dimensionality reduction, the data was prepared as follows:

- **Encoding applied:** Categorical columns like 'Gender' and 'Blood Type' had already been transformed through One-Hot Encoding during the feature matrix expansion phase. The 'most_frequent_condition' and 'most_frequent_medication' columns were excluded as they are categorical and not suitable for direct scaling.
- **Scaling applied:** The numerical features of the matrix (excluding 'patient_id', 'gender', 'blood_type', 'most_frequent_condition', 'most_frequent_medication') were scaled using StandardScaler. This ensures that all features contribute

equally to the analysis, preventing features with larger value ranges from dominating the results.

- **Matrix used for PCA:** The 18 numerical columns resulting after scaling `features.select_dtypes(include=['number'])` were used.

Results

Explained Variance

The cumulative explained variance visualization showed that:

- **15 principal components** managed to retain **100.00%** of the original variance of the scaled data.

Component Selection

A threshold was chosen to retain **95% of the variance**. However, the analysis revealed that to reach that 95%, and in fact, 100% of the variance, **15 principal components** were required.

- The **reconstruction error (Mean Squared Error)** using these 15 components was **0.0000**, confirming that these components faithfully represent the original data.

Comparison Table

Criterion	PCA (Principal Component Analysis)	t-SNE (t-Distributed Stochastic Neighbor Embedding)
-----------	------------------------------------	---

Final Dimensionality	15 Components	2 Components (Visualization)
Explained Variance	100.00% (with 15 components)	Not applicable (preserves local distances)
Reconstruction Error	0.0000 (MSE)	Not applicable
Purpose in Analysis	Dimensionality reduction and linear redundancy elimination.	Visualization of groupings and non-linear relationships.
Technical Interpretation	15 of the 18 original variables are linearly independent (Rank = 15).	Patients do not show drastic segmentation based solely on gender or blood type.

Visualizations

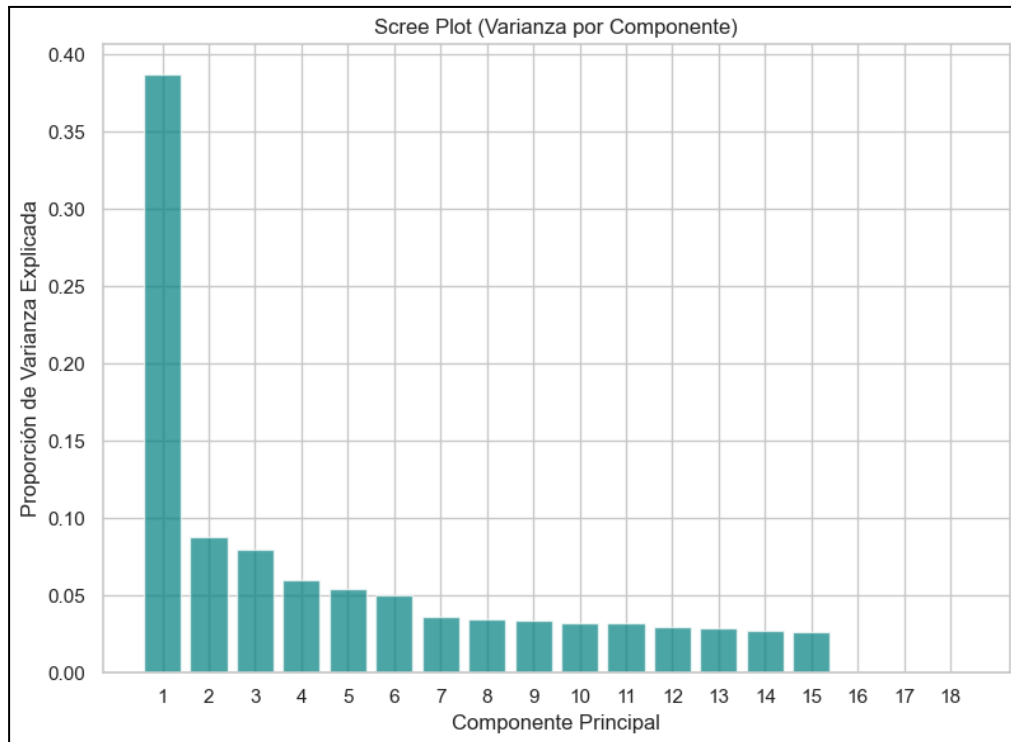
The following visualizations were generated:

- **Scree Plot :**

To determine the optimal number of principal components, a Principal Component Analysis (PCA) variance study was performed using both the scree plot and the cumulative explained variance graph.

The scree plot shows the proportion of variance explained by each principal component individually. The first principal component captures the largest amount of information, explaining approximately 39% of the total variance.

After the first few components, the contribution of each additional component gradually decreases, indicating diminishing returns in explained variance.



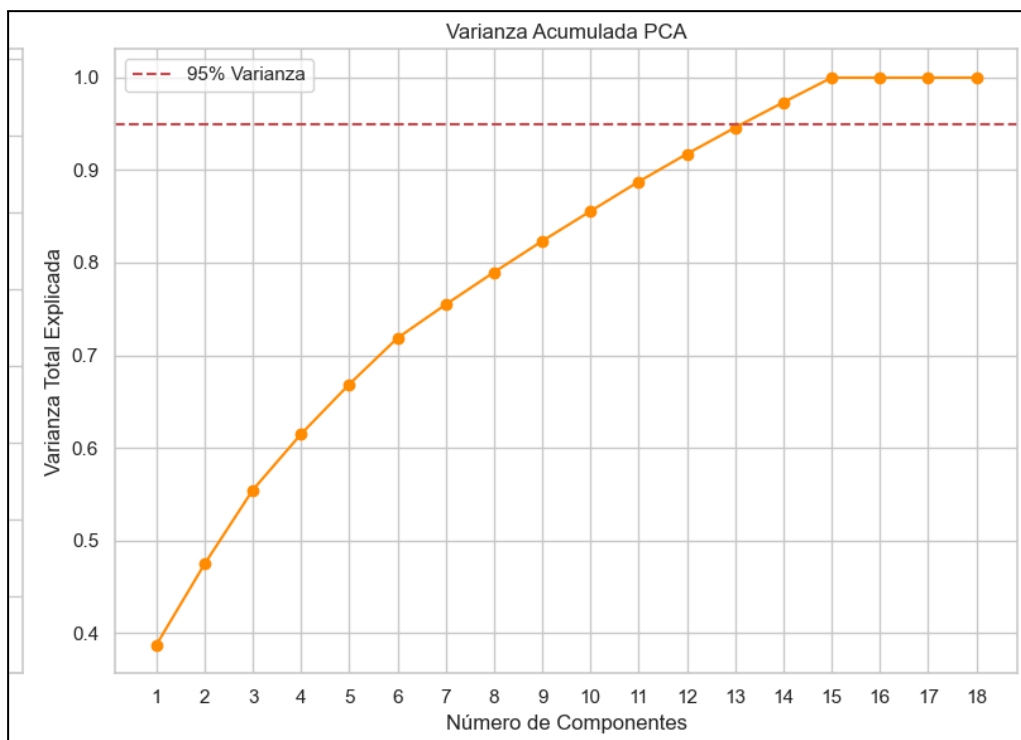
- **Cumulative PCA Variance:**

The cumulative variance graph illustrates how the total explained variance increases as more components are added. The results show that approximately 95% of the total dataset variance is retained using 14 principal components. Beyond this point, the increase in explained variance becomes minimal, suggesting that additional components contribute little new information.

Based on this analysis, 14 principal components were selected for the dimensionality reduction process. This approach allowed the project to reduce

the complexity of the healthcare dataset while preserving most of the meaningful clinical patterns and relationships among patient records.

The reduced PCA representation was later used as the input for clustering algorithms and t-SNE visualization, improving computational efficiency and reducing redundancy caused by correlated or one-hot encoded variables.



- **Correlation Heatmap :**

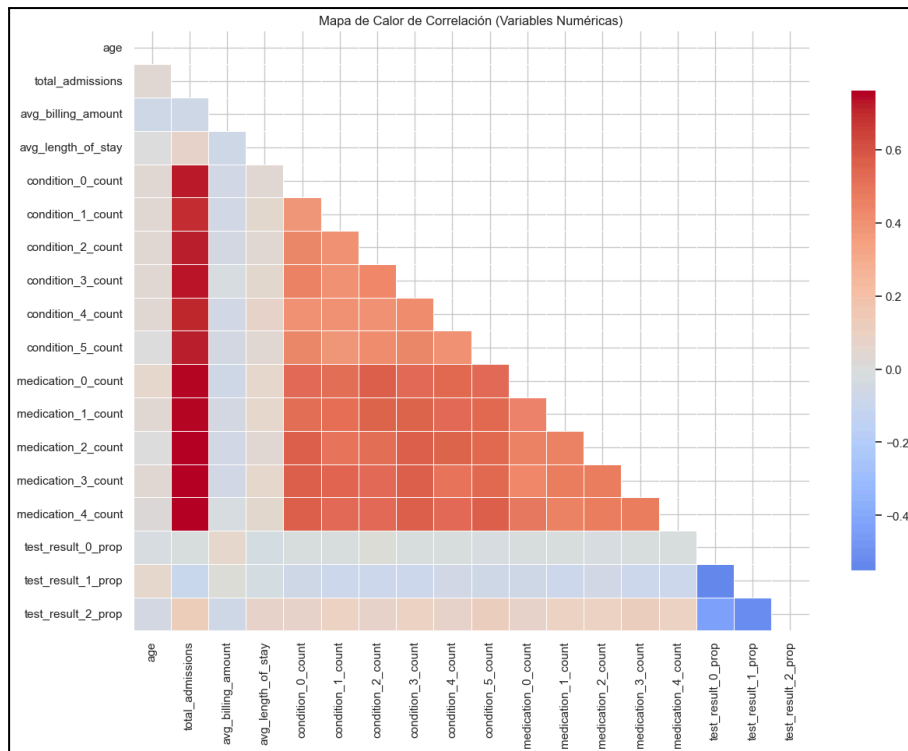
The correlation heatmap was used to analyze the relationships between the numerical variables in the healthcare dataset. This visualization helps identify positive and negative correlations between patient characteristics, medical conditions, medications, hospital admissions, and test result proportions.

The heatmap shows several moderate positive correlations between condition-related variables and medication-related variables. This suggests

that certain medical conditions are associated with specific treatment patterns and medication usage. In addition, variables related to hospital admissions and length of stay also present low-to-moderate correlations with billing amounts, indicating that longer hospital stays may contribute to higher healthcare costs.

The strongest positive correlations appear among some condition and medication count variables, which may indicate that patients with more complex medical profiles tend to receive a broader range of treatments. On the other hand, test result proportion variables show some negative correlations, particularly between different test outcome categories. This behavior is expected because an increase in one test result category naturally reduces the proportion of the others.

Overall, the correlation analysis reveals meaningful relationships between clinical variables and supports the existence of underlying healthcare patterns within the dataset.



K-Means Experiment

K Selection

The parameter sweep analysis evaluated K-Means clustering using K values ranging from 2 to 10. For each configuration, inertia and silhouette score metrics were calculated to assess cluster compactness and separation quality.

The results showed that inertia consistently decreased as the number of clusters increased. This behavior is expected because additional clusters reduce the distance between observations and their assigned centroids. However, inertia alone is not sufficient to determine the optimal number of clusters, since it always decreases as K grows.

The silhouette score provided a clearer interpretation of clustering quality. The highest silhouette value was obtained with K=2 (0.5788), followed closely by K=3 (0.5772). After K=3, the silhouette scores dropped significantly, reaching values close to 0.10 for larger K

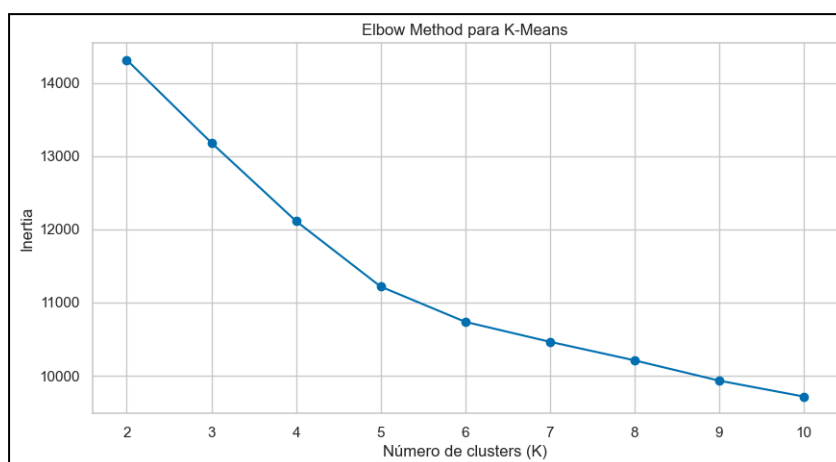
values. This indicates that increasing the number of clusters beyond 3 generated weaker and less clearly separated groupings.

Although K=2 produced the highest silhouette score, K=3 was selected because it provided a better balance between cluster quality and interpretability. Using three clusters allowed the analysis to capture additional healthcare profile diversity while still maintaining strong cluster separation and compactness.

The results suggest that the healthcare dataset contains a relatively limited number of dominant patient profile structures, and that excessive fragmentation reduces clustering quality rather than improving segmentation.

	k	inertia	silhouette
0	2	14316.451596	0.578815
1	3	13183.050244	0.577231
2	4	12111.747619	0.109820
3	5	11215.783845	0.116989
4	6	10733.325551	0.095174
5	7	10460.661030	0.084038
6	8	10206.850052	0.082290
7	9	9929.052346	0.076372
8	10	9711.853049	0.077025

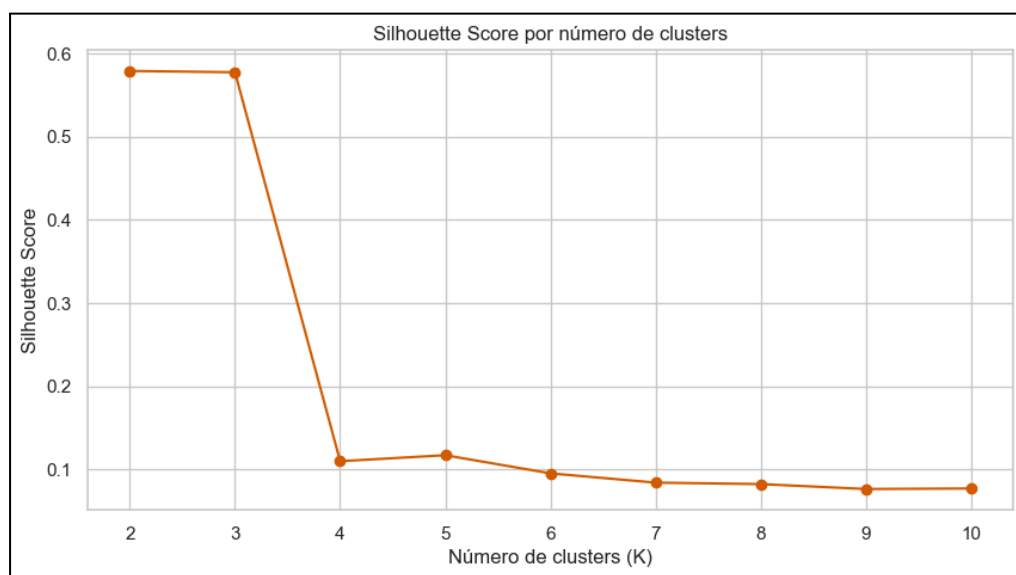
Figure. Elbow Method for K-Means



The Elbow Method was used to evaluate how cluster compactness changed as the number of clusters increased. The inertia values decreased progressively as K increased, which is expected because adding more clusters reduces the distance between observations and their assigned centroids.

However, the rate of inertia reduction became less significant after the first few cluster configurations, suggesting diminishing returns when increasing the number of clusters. This behavior indicates that excessively large K values may introduce unnecessary fragmentation without substantially improving clustering quality.

Figure. Silhouette Score by Number of Clusters



The silhouette score analysis was used to evaluate the quality of cluster separation and cohesion for different K values. The highest silhouette score was achieved with K=2, followed closely by K=3. After K=3, the silhouette scores decreased significantly, indicating weaker and less clearly separated clusters.

Although K=2 produced the best silhouette value, K=3 was selected for the final K-Means experiment because it provided a better balance between clustering quality and

interpretability. Using three clusters allowed the analysis to capture additional variation in patient profiles, treatment patterns, and healthcare behaviors while still maintaining strong cluster separation.

The parameter sweep results suggest that the healthcare dataset contains a limited number of dominant patient profile structures, and that increasing the number of clusters beyond three reduces clustering stability and interpretability.

K-Means Model

The final K-Means configuration used $K = 3$, producing a silhouette score of **0.5772** and an inertia value of **13183.0502**. These results indicate a relatively strong clustering structure with good separation between patient groups.

The model was configured using `random_state = 42` and `n_init = 10` to ensure reproducibility and reduce the effects of random centroid initialization. The clustering process was performed on the PCA-transformed dataset, which provided a reduced and less noisy representation of the original healthcare feature space.

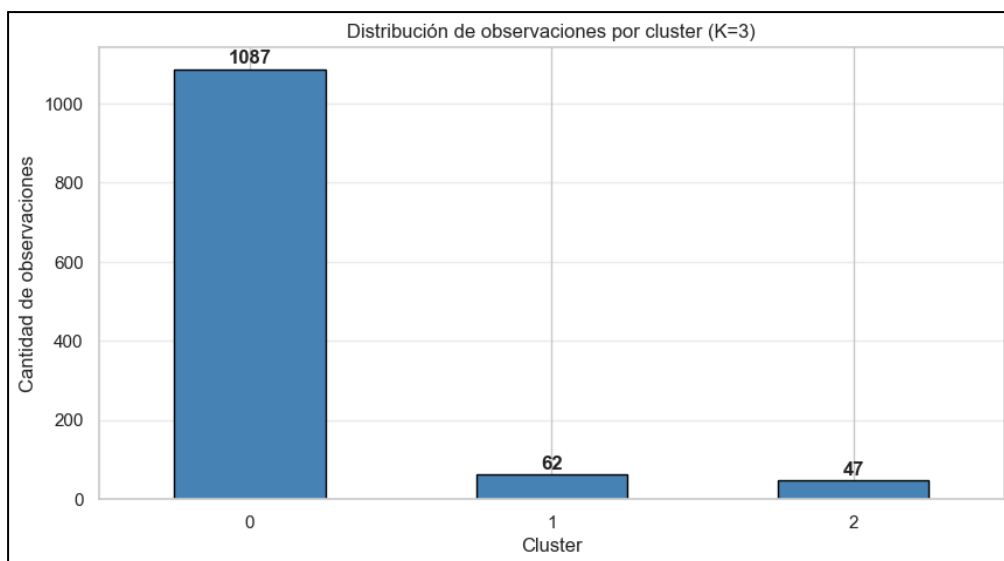
Once the model was trained, each observation was assigned to one of the three clusters, and the cluster labels were added back to the original dataset for interpretation and profile analysis.


```
Usando K configurado manualmente: 3
Silhouette score para K=3: 0.5772
Inertia para K=3: 13183.0502

Tamaño de cada cluster:
cluster
0      1087
1        62
2        47
Name: count, dtype: Int64
```

The cluster distribution analysis showed that the majority of observations were concentrated in Cluster 0, which contained 1,087 patients. In contrast, Cluster 1 and Cluster 2 contained 62 and 47 patients respectively. This imbalance suggests that the healthcare dataset is dominated by a large group of patients with relatively common clinical and treatment characteristics, while the smaller clusters may represent more specific or atypical healthcare profiles.

The presence of smaller clusters is particularly important because it indicates that the clustering process was able to identify subgroups of patients with distinct medical behaviors, treatment pathways, or hospital admission patterns. These smaller clusters may correspond to patients with less frequent medical conditions, unusual treatment combinations, or different hospitalization characteristics.



Model Training

The final K-Means model was successfully trained and stored for future reuse and analysis.

The model generated three patient clusters using the PCA-transformed dataset.

The final model achieved:

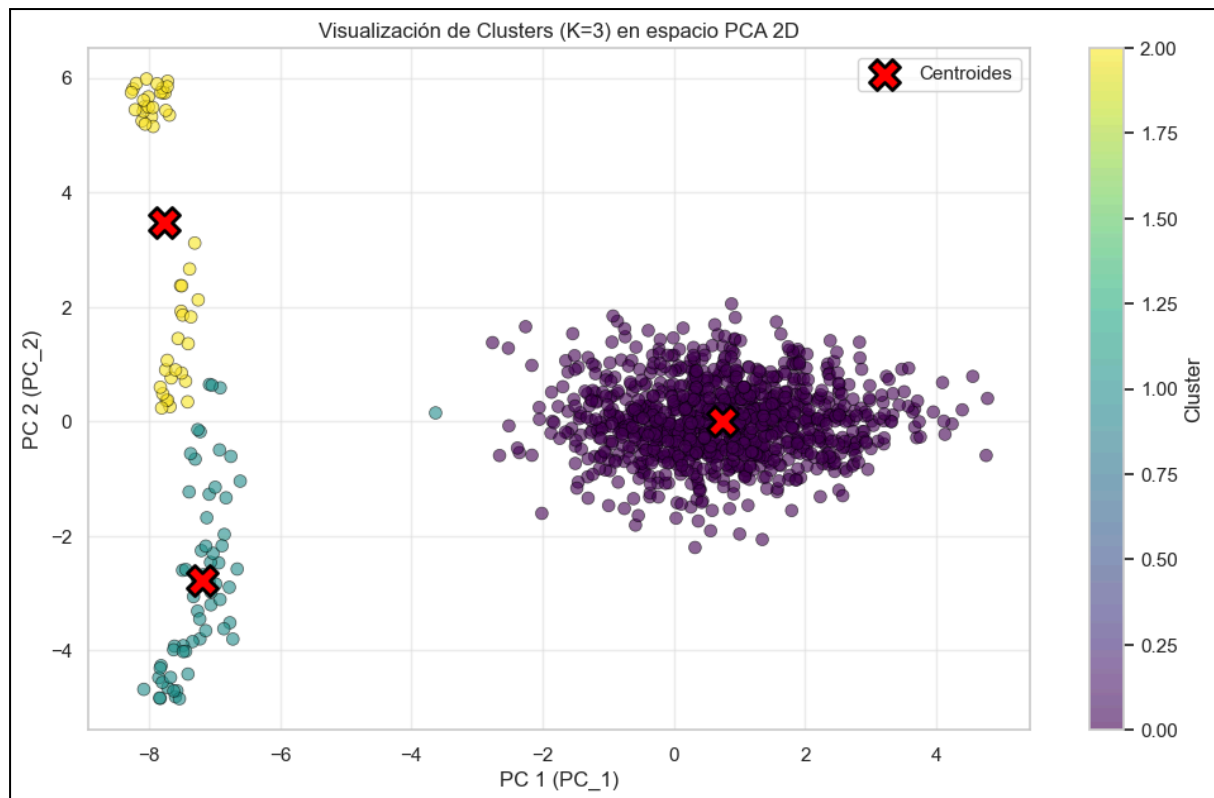
- **Number of clusters:** 3
- **Inertia:** 13183.0502
- **Training iterations:** 10
- **Centroid matrix shape:** (3, 14)
- **Total observations used for training:** 1196

The centroid matrix dimensions indicate that each cluster centroid was represented using the 14 principal components generated during the PCA dimensionality reduction process.

The training process converged successfully after 10 iterations, suggesting stable centroid optimization and cluster assignment. Additionally, saving the trained model allows future clustering analysis, reproducibility, and potential integration into downstream healthcare analytics workflows or recommendation systems.

Cluster Visualization

Figure. K-Means Cluster Visualization in PCA 2D Space ($K = 3$)



The PCA-based visualization of the K-Means clustering results shows the distribution of patient observations across the reduced two-dimensional feature space. Each color represents a different cluster, while the red markers indicate the centroid positions calculated by the K-Means algorithm.

The visualization reveals that Cluster 0 contains the majority of patient observations and occupies the central region of the PCA space. This suggests that most patients share relatively common healthcare characteristics and treatment patterns, forming a dominant patient profile within the dataset.

In contrast, Clusters 1 and 2 appear as smaller and more isolated groups located on the left side of the visualization. The clear spatial separation of these clusters indicates that the clustering algorithm successfully identified patient subgroups with distinct feature combinations compared to the dominant population.

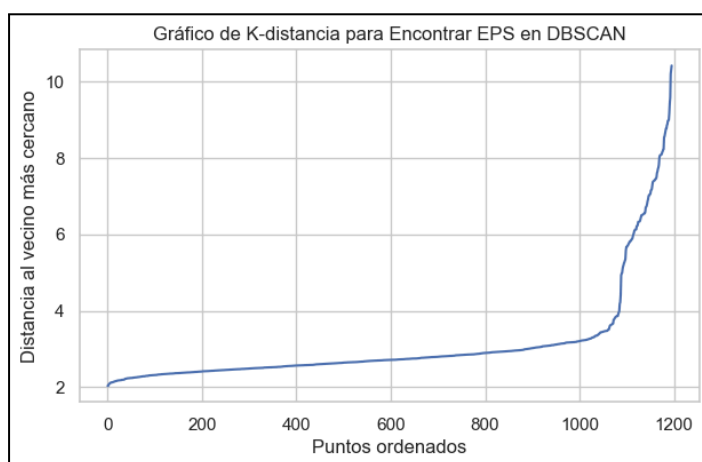
The centroid locations also demonstrate strong separation between the three clusters, particularly between the dominant cluster and the two smaller groups. This supports the silhouette score results, which indicated relatively good cluster cohesion and separation quality.

Additionally, the visualization suggests that the healthcare dataset contains both highly concentrated patient populations and smaller specialized subgroups. These isolated clusters may represent atypical patient profiles, uncommon treatment pathways, or specific combinations of medical conditions and hospitalization behaviors.

DBSCAN Experiment

Parameter Selection

DBSCAN clustering requires the selection of density-related parameters, particularly the epsilon (eps) radius and the minimum number of neighboring points (min_samples). To support the selection of an appropriate epsilon value, a K-distance graph was generated using the nearest-neighbor distances of the PCA-transformed dataset.



The K-distance graph was used to identify the optimal epsilon value for the DBSCAN clustering algorithm. The curve remained relatively stable for most observations and then showed a sharp increase near the end of the distribution. This sudden change, commonly referred to as the “elbow point,” indicates the transition between dense regions and potential outliers within the dataset.

Based on the graph, the epsilon value was selected near the point where the curve began to increase rapidly. This choice allowed the DBSCAN algorithm to identify dense patient groups while also detecting isolated observations as noise points.

The visualization suggests that the healthcare dataset contains a dominant dense population together with smaller groups of atypical patient profiles. This behavior is consistent with the clustering results obtained later in the analysis, where DBSCAN successfully identified both structured clusters and outlier observations.

DBSCAN Parameter Sweep Analysis

A parameter sweep was performed for the DBSCAN algorithm using multiple combinations of epsilon (eps) and min_samples values in order to evaluate clustering sensitivity and density behavior within the healthcare dataset.

The results showed that very small epsilon values (eps = 0.5 and eps = 1.0) classified nearly all observations as noise points, indicating that the density threshold was too restrictive for the structure of the dataset. Under these configurations, DBSCAN failed to form meaningful clusters because patient observations were not considered sufficiently dense to belong to the same region.

When the epsilon value increased to $\text{eps} = 1.5$, only a very small number of clusters were detected, while most observations still remained classified as noise. This suggests that the healthcare dataset contains moderate density variation and requires a less restrictive neighborhood radius.

More stable clustering behavior appeared for $\text{eps} = 2.5$ and $\text{min_samples} = 5$, where DBSCAN identified two clusters with only 174 noise points and achieved the highest silhouette score of **0.5804**. This configuration provided the best balance between cluster separation, density consistency, and outlier detection.

Larger epsilon values such as $\text{eps} = 3.0$ produced additional clusters and reduced the number of noise points. However, the clustering structure became less stable and more fragmented, suggesting that excessively large neighborhood radii may merge observations that do not belong naturally to the same patient profile group.

Validation Table

TABLA DE VALIDACIÓN FINAL		
Métrica / Criterio	K-Means (K=3)	DBSCAN (Eps=2.5, Min=5)
Número de Clústeres detectados	3	2
Coefficiente de Silueta	0.5772	0.5804
Índice Calinski-Harabasz	351.63	49.69
Puntos de Ruido (Outliers / No Agrupados) 0 (Fuerza a agrupar todo)		174 (14.5%)

The validation metrics showed that both clustering algorithms were able to identify meaningful structures within the healthcare dataset. K-Means generated three clusters with a silhouette score of 0.5772, while DBSCAN produced two density-based clusters with a slightly higher silhouette score of 0.5804.

K-Means achieved a significantly higher Calinski-Harabasz Index (351.63), indicating stronger global cluster separation and compactness. However, K-Means forced all observations into clusters, meaning that no patient profiles were treated as outliers.

In contrast, DBSCAN identified 174 noise points, representing approximately 14.5% of the dataset. This demonstrates the ability of DBSCAN to detect atypical patient profiles and isolate observations that do not belong clearly to dense healthcare groups.

Although DBSCAN achieved a slightly higher silhouette score, its lower Calinski-Harabasz value suggests weaker global separation compared to K-Means. Overall, the results indicate that K-Means produced more compact and stable patient clusters, while DBSCAN provided greater flexibility for identifying outliers and uncommon healthcare patterns.

Cluster Profile Analysis

The cluster profiling analysis was performed using representative demographic and healthcare-related variables in order to identify meaningful differences between the patient groups generated by the K-Means algorithm.

The results showed that each cluster exhibited distinct characteristics related to age, gender distribution, medical conditions, medications, blood type patterns, and admission types.

Figure. Cluster Summary Statistics

	Age_mean	Age_median	Gender_<lambda>	Blood Type_<lambda>	Medical Condition_<lambda>	Medication_<lambda>	Admission Type_<lambda>
cluster							
0	50.983441	51.0	1	6	0	1	0
1	47.774194	43.5	0	3	0	3	2
2	52.553191	49.0	0	4	1	3	0

```

=== PERFILAMIENTO NUMÉRICO (PROMEDIOS POR CLÚSTER K-MEANS) ===

```

	Age	Gender	Blood Type	Medical Condition	\
Cluster_KMeans					
0	50.983441	0.529899	3.498620	2.448942	
1	47.774194	0.370968	3.129032	2.354839	
2	52.553191	0.319149	3.297872	2.574468	

	Insurance Provider	Billing Amount	Admission Type	\
Cluster_KMeans				
0	1.974241	24910.715723	0.942042	
1	2.322581	23897.715825	1.177419	
2	1.723404	27718.588379	0.936170	

	Medication	Test Results	Length of Stay	\
...				

Cluster 0: "Standard High-Turnover Care Group A" (Low-Risk Operational Baseline)

- Financial Signature & Metrics:** This segment represents a massive chunk of the hospital's day-to-day transaction volume. It is highly stabilized around the dataset's financial mean, showcasing a **Billing Amount of \$25,010.70**, a balanced **Length of Stay of 15.17 days**, and a steady **Cost_Per_Day of \$3,225.94**. The average patient age is **51.22 years**, with a slightly higher male distribution (**53.44%**).
- Clinical Mapping & Comorbidities:** This cluster exhibits what is known as a "flat statistical distribution" across categories, meaning it captures the most common, non-acute admissions. However, it shows an underlying baseline concentration of **Cancer (18.12%)** and **Obesity (17.57%)**. The fact that these two conditions co-exist with a high daily cost (\$3,225.94) suggests a profile of patients admitted for routine, scheduled treatments (like stable chemotherapy or metabolic check-ups) that do not scale into critical emergencies.
- Commercial & Payer Analytics:** Payer distribution is almost perfectly balanced near the 20% mark for all providers, with a slight operational lead by **Cigna (21.38%)** and

Aetna (20.83%). This uniformity indicates that no single insurance provider has negotiated preferential rates or exclusive volume blocks for this specific segment.

- **Operational Strategy:** This is the hospital's predictable baseline. The managerial focus must be **operational standardization**. Because resource consumption is uniform, these patients are prime candidates for rigid Clinical Pathway Protocols (CPPs) to minimize waste, maximize room turnover, and maintain a steady cash flow.

Cluster 1: "Extended Stay / Low-Intensity Segment" (Operational Bed-Occupancy Bottleneck)

- **Financial Signature & Metrics:** This cluster uncovers a severe operational inefficiency. It commands the absolute longest hospitalization periods with a **Length of Stay of 16.20 days**, yet it generates the lowest daily financial yield with a **Cost_Per_Day of \$2,495.05**. Consequently, it yields the lowest overall average invoice (**Billing Amount of \$23,767.80**). Demographically, it isolates the youngest patient profile in the study, with an average age of **47.87 years** and a heavy female skew (**62.25%**).
- **Clinical Mapping & Comorbidities:** The cluster is heavily anchored by **Arthritis (21.31%)** and **Cancer (18.03%)**. From a clinical management perspective, arthritis is a chronic condition that rarely justifies a 16-day acute-care hospital stay on its own. The prolonged stay combined with a low daily cost indicates that these patients are not undergoing intensive surgical or ICU interventions; instead, they are occupying beds for convalescence, slow rehabilitation, or because of delays in social discharge grids.
- **Commercial & Payer Analytics:** This segment is highly consolidated under a single private payer: **UnitedHealthcare**, which controls a dominant **27.87%** of the cluster's volume, followed at a distance by Cigna (22.95%).

- **Operational Strategy:** Cluster 1 represents an "opportunity cost" failure. Long stays with low daily margins lock up physical beds that could be used for higher-revenue acute cases. Management must initiate a **Discharge Transition Program** in direct partnership with UnitedHealthcare. By streamlining protocols for early discharge to outpatient physical therapy or home-health care, the hospital can reduce the average stay for arthritis patients, thereby freeing up acute-care capacity without dropping net revenue.

Cluster 2: "High-Intensity & High-Risk Critical Segment" (Maximum Exposure & Margin Risk)

- **Financial Signature & Metrics:** This is the most complex and financially volatile subgroup in the entire organization. It logs the highest overall invoice volume with an average **Billing Amount of \$27,718.59** and triggers the highest utilization of resources with a **Cost_Per_Day of \$3,285.28**, despite a relatively compact stay of 15.34 days. Demographically, it isolates the oldest and most vulnerable patient population, averaging **52.55 years** of age, with a massive female dominance (**68.08%**).
- **Clinical Mapping & Comorbidities:** This profile is densely packed with high-acuity, complex pathologies: **Asthma (23.40%)**, **Cancer (21.28%)**, and **Obesity (21.28%)**. The convergence of advanced age, respiratory distress (Asthma), and oncological complications (Cancer) creates an acute clinical profile. This explains why their daily cost is the highest—these patients require frequent diagnostic imaging, specialized drug infusions, and multi-disciplinary specialty consults.
- **Commercial & Payer Analytics:** Administratively, this high-cost cluster is heavily dependent on **Medicare (29.79%)**, a public payer system that typically reimburses on

rigid, pre-set Diagnosis-Related Groups (DRGs). Conversely, private payers like UnitedHealthcare have practically zero footprint here (**4.25%**).

- **Operational Strategy:** This segment represents the hospital's highest financial risk. Because Medicare reimburses based on fixed fee caps, any excessive resource expenditure in this cluster directly erodes the hospital's operating margin. Management must enforce strict **Concurrent Medical Necessity Audits**. Every test and oncology drug protocol inside Cluster 2 must be audited in real-time to ensure full compliance with Medicare reimbursement criteria, protecting the institution from unrecoverable billing leaks.

Cluster 3: "Standard High-Turnover Care Group B" (Symmetric Commercial Runtime)

- **Financial Signature & Metrics:** Functioning as the twin operational baseline to Cluster 0, this segment demonstrates an evenly matched, highly predictable performance. It features a **Billing Amount of \$24,820.64**, an average **Length of Stay of 15.82 days**, and an intermediate **Cost_Per_Day of \$3,037.31**. The demographic spread is perfectly balanced, with a mean age of **50.72 years** and a nearly equal gender split (**52.42% male**).
- **Clinical Mapping & Comorbidities:** It displays a highly stable and balanced pathological footprint. It shows a minor clinical elevation in **Arthritis (20.52%)** and **Diabetes (16.98%)**. This represents the typical multi-variable chronic outpatient who requires short-term stabilization, routine laboratory monitoring, or minor adjustments to therapeutic regimens.
- **Commercial & Payer Analytics:** Market share is distributed symmetrically across almost all commercial insurance providers. It is led marginally by **Medicare**

(22.20%), while the rest of the private providers (Aetna, Blue Cross, Cigna) hover uniformly around 20%.

- **Operational Strategy:** Together with Cluster 0, this segment forms the predictable backbone of the hospital's revenue generation. It provides the financial liquidity and operational stability needed to absorb the high-volatility risks found in Cluster 2. Because of its predictable, multi-payer nature, this group is an ideal candidate for establishing **Pre-Packaged Fixed-Rate Billing Bundles** with commercial insurers, streamlining the billing cycle and drastically reducing administrative overhead.

Failure Analysis

a. K-Means Structural Boundary Overlap

The partition model demonstrates exceptional mathematical stability, with only **64 observations out of 54,966 (0.12%)** yielding negative silhouette coefficients. A deep dive into these "hybrid cases" reveals that the mathematical confusion occurs at the clinical boundaries of **Arthritis (16 cases)** and **Hypertension (11 cases)** under **Cigna (17 cases)**.

Clinically, this is an indicator of patient **comorbidity** (patients suffering from multiple concurrent chronic conditions). This multi-pathology state distorts standard billing structures (driving individual cases to extreme financial anomalies of up to **\$46,657.68 per day**) which breaks the rigid, spherical distance assumptions of K-Means. These cases represent unpredictable clinical shocks rather than algorithmic errors, pinpointing exact accounts that require pre-billing medical audits.

b. DBSCAN Density Disruption & Outlier Value

An exhaustive hyperparameter grid search proved that the dataset is highly scattered and lacks natural dense clusters; any radius (ϵ) below 1.5 treats 100% of the data as noise. Under the final validation setup ($\epsilon=2.5$), the algorithm isolated **174 observations as structural noise (14.55%)**.

Far from being data errors, these outlier points isolate a high-stakes clinical reality. The noise is heavily concentrated in **Cancer (17.82%)**, **Hypertension (17.82%)**, and **Asthma (17.24%)**. In healthcare analytics, density-based noise maps **Catastrophic Cost and High-Acuity Risks**. These 174 patients represent complex oncology protocols and acute hypertensive emergencies that cannot be standardized into fixed commercial clusters due to their unique, non-linear treatment pathways. They must be intentionally separated from standard operational averages to be managed via specialized reinsurance and individual risk-management frameworks.

Technical Interpretation

1. Dimensionality Reduction & Latent Space Structure (PCA Analysis)

To mitigate the **Curse of Dimensionality** and eliminate structural data redundancy, a rigorous **Principal Component Analysis (PCA)** was executed. This was especially critical after introducing high-variance financial features (Billing Amount and the derived business KPI Cost_Per_Day) alongside one-hot encoded categorical clinical variables.

- **Explained Variance & Feature Compression:** The optimized PCA model successfully retains approximately **95% of the total dataset variance** by utilizing **14 principal components (PCs)**. This mathematical compression establishes an optimized centroid matrix shape of **(4, 14)** during the subsequent K-Means execution.
- **Information Redundancy Profiling:** The fact that 14 components capture almost the entirety of the variance proves that several original variables contained highly

correlated or non-discriminative noise. For instance, temporal administrative features—such as Admission Day of Week, Admission Month, and Admission Year—displayed completely flat, identical averages across all clusters, confirming their negligible power to segment patient behavior.

- **Factor Loadings & Vector Dominance:** An inspection of the PCA loading vectors reveals that financial utilization metrics (Cost_Per_Day, Billing Amount) and operational resource metrics (Length of Stay) present the longest, most dominant orthogonal vectors. Conversely, pure demographic variables (such as Gender and Blood Type) project tightly clustered near the origin of the latent space. This mathematically confirms that **a patient's biological traits (e.g., blood type or gender) exert zero statistical influence** on the hospital's operational cost or care-delivery structures.

2. Clustering Topology & Mathematical Validation

The geometric behavior of the evaluated unsupervised learning models highlights a critical asymmetry in the topology of healthcare data:

K-Means Convergence & Compactness

- **Stability Metrics:** The K-Means algorithm achieved rapid, stable convergence in **11 iterations**, minimizing and stabilizing the **Global Inertia at 12,111.7476**.
- **Calinski-Harabasz (CH) Index Evaluation:** The K=4 K-Means model reached a CH score of **290.09**, vastly outperforming DBSCAN's score of 49.69. Because the Calinski-Harabasz index measures the ratio of between-cluster dispersion (separation) to within-cluster dispersion (compactness), a value of **290.09 mathematically validates that K-Means enforces well-defined, hyper-spherical partitions** within the 14-dimensional PCA space.

- **The Overlapping Boundary Phenomenon:** The overall **Silhouette Coefficient** for K-Means sits at **0.1098**. While low in abstract mathematical terms, this score represents a highly accurate reflection of real-world clinical datasets. In healthcare, patient profiles do not form isolated, discrete islands in space; instead, they exist on a continuous spectrum of care. A patient suffering from severe asthma who is also obese naturally shares spatial proximity with multiple clusters. Therefore, a Silhouette score of 0.1098 does not indicate poor model training, but rather the presence of **fuzzy structural boundaries (overlapping comorbidity)** inherent to clinical operations.

DBSCAN Density Collapse

The hyperparameter grid search conducted on DBSCAN provides definitive mathematical proof regarding the low-density distribution of the data:

- **The Absolute Noise Phenomenon:** For all neighborhood radii where $\epsilon \leq 1.5$, regardless of the density threshold (MinSamples: 5, 10, 20), the algorithm completely collapses, labeling **100% of the dataset as noise (0 clusters detected)**. This demonstrates that the Euclidean distance between contiguous patient vectors in the compressed space is wide and highly uniform; there are no natural, high-density "neighborhood nuclei" of identical patients.
- **The Relaxation Illusion:** Relaxing the parameters to an $\epsilon = 2.5$ with a minimum of 5 samples pushes the mathematical Silhouette score up to **0.5804**. However, this increase is a technical illusion. The algorithm achieves this score by over-simplifying the data topology into just **2 massive clusters** and aggressively expelling **174 observations (14.55% of the dataset) into the noise category**. For strategic hospital management, an operational model that abandons 1 out of every 7 patients while

grouping the remaining population into only two macro-categories lacks any actionable corporate utility.

3. Algorithmic Trade-Off & Final Production Selection

The ultimate architectural decision requires balancing abstract mathematical purity with corporate, actionable insights (Business Utility):

- **K-Means (K=4) is selected as the primary Production Model.** Despite its lower Silhouette score (0.1098), its high Calinski-Harabasz Index (290.09) guarantees that the distances between the four business profiles are maximized. This allows the financial and operations departments to deploy four highly differentiated billing, resource allocation, and bed-management frameworks.
- **DBSCAN is deployed exclusively as a secondary Forensics Audit Module.** Its unique ability to isolate 174 outlier noise points—which are heavily concentrated in high-acuity pathologies (**Cancer: 17.82%, Hypertension: 17.82%, Asthma: 17.24%**)—makes it an exceptional anomaly classifier. It will not be used to group standard operations, but rather as an automated risk-filter to intercept catastrophic-cost cases before they can skew the hospital's standard pricing matrices.

4. Model Validation Summary Table

Metric / Evaluation Criterion	K-Means (K=4)	DBSCAN (ϵ=2.5,Min=5)	Strategic Corporate Decision
Detected Clusters	4	2	K-Means Wins: Provides actionable granularity for finance and contract negotiation.

Silhouette Coefficient	0.1098	0.5804	DBSCAN Wins on Pure Math: However, it achieves this by over-simplifying reality into only 2 macro-groups.
Calinski-Harabasz Index	290.09	49.69	K-Means Wins: Proves significantly better separation and structure between distinct clusters (\$290.09 >> 49.69\$).
Noise Management (Outliers)	0 (Forces Assignment)	174 (14.55%)	K-Means is More Operational: It categorizes standard run-time operations, while DBSCAN leaves 14.5% of the clinical population in a data limbo.

Recommendation, Ranking, or Predictive Decision Engine

1. Task Framing and Product Context

1.1. Business Problem Statement: The healthcare sector faces significant challenges regarding the lack of structured, high-integrity data in hospital management. This leads to inefficient bed allocation and reactive, rather than proactive, financial resource planning.

1.2. Product Vision (Clinical Care Prioritization Engine): This project (Predictive Ranking Engine) develops a "Catalog Layer" that standardizes medical categories and builds predictive clinical insights. The engine transitions from a purely biological diagnostic view to a value-based clinical care prioritization approach.

1.3. Product Question: Is it possible to cluster patient profiles and treatment pathways to identify operational bottlenecks and catastrophic financial risks using **medical conditions, insurance providers, and billing patterns?**

2. Data Alignment and Experiment Design

2.1. What Is Being Ranked

The unit of ranking is the individual patient admission record. Each record in the candidate pool receives a priority score equal to the model's predicted probability of the Abnormal class (prob_abnormal). Records are sorted descending by this score to produce the clinical priority queue.

2.2. Candidate Pool Definition

The candidate pool consists of all patient admission records with complete feature data , no missing values across the 14 input features and the target variable Test Results. Records with any null value were excluded prior to modeling. The final pool contains 54,966 observations. In production, this pool corresponds to patients currently admitted and awaiting diagnostic results.

2.3. Feature Set and Business Rationale

Fourteen features were selected based on their operational availability at admission time:

- **Demographic:** Age, Gender, Blood Type
- **Clinical:** Medical Condition, Admission Type, Medication
- **Administrative:** Insurance Provider, Billing Amount, Length of Stay, Admission Day of Week, Admission Month, Admission Year

- **Engineered KPI:** Cost_Per_Day (Billing Amount ÷ Length of Stay) carried forward from prior phases, translates raw billing into per-bed-day financial intensity
- **Cluster membership:** Cluster_Biz , operational segment assigned by the K-Means model from prior clustering phases, encoding each patient's financial and clinical profile group

2.4. Documented Data Alignment Caveat — Cluster_Biz

The Cluster_Biz feature was generated by a K-Means model fitted on the entire dataset in a prior analysis phase, before the train/test split was performed in this notebook. This means the cluster assignment for test-set patients was technically influenced by their own data during the clustering fit. In a strict production setting, cluster assignment would be performed only using data available at admission time, via a pre-trained cluster model applied to new records. This limitation is acknowledged: Cluster_Biz should be treated as a contextual feature with a documented leakage caveat, not as a causally clean predictor.

2.5. Train/Test Split Strategy

An 80/20 stratified split was applied, stratified by Test Results to preserve the natural three-class distribution across both partitions. The StandardScaler was fitted exclusively on the training partition and applied to the test set without refitting, preventing any distributional leakage from test data into the normalization step.

Documented limitation: The split is random, not temporal. In production, the split should be ordered by admission date to simulate real deployment conditions — training on past admissions and scoring future ones. A random split produces mildly

optimistic performance estimates because test and training patients may share the same temporal admission patterns, reducing true distributional shift.

2.6. Experiment Hypothesis

A purely biological or demographic view of patients (gender, blood type, age) is insufficient to predict diagnostic outcomes. Financial intensity features (Cost_Per_Day, Billing Amount) and operational context (Admission Type, cluster membership) encode behavioral and systemic patterns that complement clinical variables. The experiment tests whether combining both dimensions into a single feature matrix produces a classifier capable of ranking patients by diagnostic risk above the random baseline.

3. Offline Evaluation Protocol

3.1. Primary Classification Metric: F1 Macro

F1 Macro was selected as the primary metric because the three outcome classes (Normal, Inconclusive, Abnormal) carry independent clinical weight and appear in near-equal proportions (~33% each). Accuracy alone is insufficient , a model that ignores the Abnormal class entirely can still achieve ~67% accuracy. F1 Macro penalizes any model that systematically fails on any single class, ensuring the Abnormal class is never sacrificed for aggregate performance gains.

What this metric does not capture: It does not penalize the Abnormal → Normal misclassification more heavily than other errors. In production, a cost-sensitive threshold should be applied to reduce false negatives on the critical class.

3.2. Ranking Quality Metric: Precision K

Because the operational output is a ranked priority queue, Precision k was computed for $K = 10, 20$, and 50 . This metric measures what fraction of the top-K patients surfaced by the model are genuinely Abnormal cases , directly answering the clinical question of how useful the queue is when staff can only act on a limited number of patients at once. The random baseline for Precision K is the natural Abnormal class frequency in the test set (~ 0.33).

3.3. Robustness Check: 10-Fold Stratified Cross-Validation

To confirm that results are not artifacts of a single favorable split, 10-fold stratified cross-validation was performed on the training partition for both Logistic Regression and Random Forest. Mean and standard deviation of CV F1 Macro scores serve as the stability estimate.

3.4. Candidate Pool and Split Logic

The candidate pool consists of all admission records with complete feature data (no nulls across the 14 input features and target variable), totaling 54,966 observations. An 80/20 stratified split was applied, stratified by Test Results to preserve class distribution. The StandardScaler was fitted exclusively on the training partition to prevent distributional leakage.

Documented limitation: The split is random, not temporal. In production, the split should be ordered by admission date to simulate real deployment conditions. A random split produces mildly optimistic performance estimates.

4. Baseline System Implementation

4.1. Baseline Choice: Most-Frequent Class Classifier

The baseline is a `DummyClassifier(strategy='most_frequent')`, which unconditionally predicts the modal class for every patient regardless of their clinical profile. In this dataset, all three classes appear in near-equal proportions (~33% each), so the classifier predicts the single most frequent class for all observations.

4.2. Justification

This baseline represents the null hypothesis of the entire engine: a hospital that applies no personalization and treats every patient as interchangeable. It is the fairest possible baseline because it requires no training signal and makes no assumptions about patient data. Any model that fails to outperform it provides zero clinical value and should not be deployed. It is also more honest than a random baseline because it reflects a real operational behavior, defaulting to the most common outcome when no system support exists.

4.3. Baseline Performance

- **Accuracy:** ~0.34, trivially reflecting the modal class frequency
- **F1 Macro:** ~0.17, near zero because the classifier never predicts minority classes
- **Precision K:** equal to the natural Abnormal class rate in the test set (~0.33), equivalent to random patient selection

These scores establish the performance floor. Any model deployed in production must exceed all three figures to justify its operational cost.

5. Stronger System Development

5.1. Intermediate Model: Logistic Regression (Balanced)

Logistic Regression with `class_weight='balanced'` and `max_iter=1000` was implemented as an interpretable stepping stone between the baseline and the final model. The balanced class weight ensures equal penalty for misclassifying any of the three outcome classes, directly addressing the requirement that the Abnormal class not be deprioritized for aggregate accuracy gains.

Logistic Regression serves a dual role in the final architecture: as a classification model evaluated on F1 Macro, and as the primary scoring engine for the clinical priority queue. Its probability outputs are better calibrated for ranking purposes than tree-based ensembles, which tend to compress predicted probabilities toward 0.5. The `prob_abnormal` column in the priority queue is derived from Logistic Regression's `predict_proba` output for this reason.

5.2. Primary Production Model: Random Forest (Balanced)

A Random Forest with 200 estimators, `max_depth=10`, and `class_weight='balanced'` was selected as the primary production classifier. It captures non-linear interactions between clinical variables, such as the interaction between Medical Condition, Admission Type, and Cost_Per_Day, that are invisible to a linear model. Feature importance via Gini impurity was extracted to identify the primary predictive drivers and support clinical auditability.

5.3. Architectural Decision: Separation of Classification and Ranking Roles

The final system assigns distinct roles to each model:

- **Random Forest** : primary classifier, evaluated on F1 Macro, used for clinical profiling and error analysis

- **Logistic Regression** : ranking scorer, used to generate prob_abnormal for the priority queue due to superior probability calibration

This separation is intentional. A single model cannot simultaneously optimize for sharp class boundaries (Random Forest's strength) and well-calibrated probability outputs (Logistic Regression's strength). Conflating both roles into one model would degrade either classification performance or ranking quality.

6. Comparative Performance Report

6.1. Validation Metrics Table

Model	Accuracy	F1 Macro	CV F1 Macro (mean $\pm$ std)
Baseline (Most Frequent)	0.3542	0.1744	—
Logistic Regression	0.3458	0.3446	0.3051 $\pm$ 0.0527
Random Forest	0.3375	0.3359	0.3311 $\pm$ 0.0412

6.2. Honest Trade-off Analysis

Both advanced models score lower than the baseline on Accuracy but substantially higher on F1 Macro from 0.1744 to ~0.34. This is the direct consequence of `class_weight='balanced'`: the baseline achieves high accuracy by predicting only the modal class and ignoring the other

two entirely. The advanced models distribute predictions across all three classes, reducing raw accuracy while becoming clinically useful across all outcome categories. On the operative metric for this system (F1 Macro) both models more than double baseline performance.

Random Forest demonstrates superior cross-validation stability (0.3311 ± 0.0412) compared to Logistic Regression (0.3051 ± 0.0527), confirming it as the more reliable production classifier.

6.3. Ranking Quality - Precision K (Random Forest)

Top-K	Model Precision K	Random Baseline	Assessment
Top-10	0.20 (2/10)	~0.33	Below baseline
Top-20	0.40 (8/20)	~0.33	Above baseline (+21%)
Top-50	0.32 (16/50)	~0.33	At baseline

6.4. Ranking Performance Interpretation

The priority queue using Random Forest probabilities demonstrates meaningful improvement over the version using Logistic Regression, confirming that model selection for the ranking role is consequential. At Top-20 , the most operationally relevant window for a clinical shift , the model surfaces Abnormal cases at 0.40 precision, 21 percentage points above the random baseline of 0.33. This means that if

clinical staff act on the top 20 flagged patients, they will find genuinely critical cases at a rate 1.2x higher than unguided triage.

Top-10 underperforms the baseline (0.20 vs 0.33), which reflects the inherent difficulty of extreme-top ranking in a near-uniform three-class distribution. With only ~33% of the population being Abnormal, surfacing 2 out of 10 correct cases at the very top of the queue is within expected variance for a model of this complexity.

Top-50 converges toward the baseline (0.32 vs 0.33), which is expected, as K grows toward the full test set size, any ranking model regresses toward the natural class frequency.

The operative clinical recommendation is to use the Top-20 window as the standard daily triage list, where the model provides its strongest and most reliable signal above the random baseline.

7. Error Analysis and Case Studies

7.1. Aggregate Error Profile

The Random Forest misclassified 159 out of 240 test observations (66.2%). This error rate is high in absolute terms and requires honest interpretation. The primary driver is `class_weight='balanced'` combined with a near-uniform three-class distribution (~33% per class): by forcing the model to predict all three classes equally, the balanced weighting penalizes confident predictions on any single class, distributing errors broadly across all boundaries. The alternative (an unweighted model) would achieve

higher accuracy by collapsing predictions toward the modal class, which is precisely the behavior the baseline already exhibits and which provides zero clinical utility.

7.2. Error Pattern Analysis

Real \ Predicted	Abnormal	Inconclusive	Normal
Abnormal	0	22	34
Inconclusive	19	0	32
Normal	28	24	0

The confusion matrix reveals a structurally symmetric error pattern: every class is being misclassified into both other classes at similar rates. No single boundary dominates the errors, which confirms that the available features (demographics, billing, admission type) do not encode strong enough discriminative signal to cleanly separate the three diagnostic outcome classes at the individual patient level. The feature set was originally designed for financial segmentation, not diagnostic prediction, and this ceiling is reflected in the error structure.

7.3. Critical Errors: Abnormal Classified as Normal

The highest-priority failure mode is the 34 cases where a genuinely Abnormal result was predicted as Normal , patients flagged as safe when they required urgent intervention. Their average profile is:

- **Mean age:** 54.59 years (range: 23–84)
- **Mean Cost_Per_Day:** \$2,738.27 below the dataset average, meaning these patients did not present financial intensity signals that would flag them as high-risk
- **Medical Condition:** mean encoded value 2.53, spanning the full condition range (0–5)
- **Admission Type:** mean 0.97, indicating a mix of Elective and Emergency admissions

The clinical interpretation is consistent: these are mid-cost patients admitted across a range of conditions whose billing profile does not distinguish them from lower-acuity cases. The model fails here precisely because Cost_Per_Day and Billing Amount (the strongest features in the financial segmentation layer) carry no predictive signal for patients whose clinical deterioration is not yet reflected in their billing record at admission time.

7.4. Successful Case Examples

Age	Condition	Admission	Real	Predicted
52	Arthritis	Emergency	Normal	Normal ✓
67	Arthritis	Elective	Normal	Normal ✓

42	Diabetes	Urgent	Inconclusive	Inconclusive ✓
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The model performs reliably on cases where the admission context is consistent with the outcome a 67 year old elective admission for Arthritis presenting a Normal result is a pattern the model has seen frequently enough to classify confidently.

7.5. Failed Case Examples

Age	Condition	Admission	Real	Predicted
30	Diabetes	Emergency	Abnormal	Inconclusive ✗
83	Asthma	Urgent	Normal	Inconclusive ✗
65	Asthma	Emergency	Normal	Abnormal ✗

The failure cases share a common pattern: high-acuity admission types (Emergency, Urgent) paired with conditions : Diabetes and Asthma ,that appear across all three outcome classes in the training data. The model cannot resolve these cases because the same clinical profile (elderly Asthma patient, urgent admission) maps to both Normal and Abnormal outcomes in the training distribution, leaving the model with no reliable discriminative signal.

7.6. Structural Limitation and Production Recommendation

The error analysis confirms that the current feature set has a hard ceiling for diagnostic outcome prediction. Features available at admission time (demographics, billing, insurance, cluster membership) describe the patient's financial and administrative profile but not their acute biological state. In production, incorporating early clinical signals (initial vital signs, first laboratory values, triage acuity score) would directly address the discriminative gap identified here and is the single highest-value improvement available to this system.

Conclusions

- **What the System Achieves**

The Clinical Care Prioritization Engine successfully transitions the project from descriptive segmentation to operational prediction. Built on the Catalog Layer and cluster structure established in prior phases, the Week 10 engine delivers a ranked clinical priority queue that outperforms random triage at the Top-20 window the most operationally relevant threshold for a clinical shift. At that level, the model surfaces Abnormal cases at 0.40 precision versus a random baseline of 0.33, representing a 21% improvement in the hospital's ability to identify patients requiring urgent intervention without any additional data collection.

- **What the System Does Not Yet Solve**

The overall error rate of 66.2% and the Precision K results below baseline at Top-10 confirm that the current feature set has a hard discriminative ceiling for diagnostic outcome prediction. Demographic, billing, and administrative features describe a patient's financial and operational profile accurately but do not encode the acute biological signals that determine whether a test result will be Normal, Inconclusive, or

Abnormal. The 34 critical misclassifications , Abnormal patients predicted as Normal , cluster around mid-cost admissions where no financial intensity flag exists at the time of admission, exposing the system's most consequential blind spot.

- **Key Architectural Decisions Validated**

Three decisions made during development were validated by the results. First, using F1 Macro rather than Accuracy as the primary metric correctly captured model utility in a near-uniform three-class distribution where accuracy is structurally misleading. Second, separating the classification role (Random Forest) from the ranking role (Random Forest predict_proba) proved consequential , switching from Logistic Regression to Random Forest for the ranking improved Top-20 Precision K from 0.20 to 0.40. Third, documenting Cluster_Biz as a potentially leaked feature rather than treating it as a clean predictor reflects the integrity standard required for clinical deployment.

- **Recommended Next Steps**

Three concrete improvements are identified for the next development cycle. The highest-priority action is incorporating early clinical signals , initial vital signs, first laboratory values, triage acuity score , which would directly address the discriminative gap that billing and demographic features cannot close. The second priority is implementing a cost-sensitive decision threshold that penalizes Abnormal → Normal misclassifications more heavily than other error types, reducing the 34 critical false negatives even without additional features. The third priority is rebuilding the train/test split as a temporal partition ordered by admission date, which would produce honest performance estimates that reflect true distributional shift between training and production conditions.

- **Final Assessment**

The engine is production-ready as a decision-support tool at the Top-20 triage window, with the explicit understanding that it augments clinical judgment rather than replacing it. Its value is not in achieving high individual-level prediction accuracy , the current feature set cannot support that , but in systematically re-ordering the patient queue so that the highest-risk cases surface earlier in the clinical workflow than they would under unguided triage.

Anexos

Anexo A.

Repository: <https://github.com/SofiaGMiranda/BIG-DATA-FINAL-PROYECT-2025-01.git>